

CARDIFF WEATHER CENTRE, United Kingdom

WMO: 037170

Lat: 51.483N

Lon: 3.183W

Elev: 52

StdP: 100.70

Time Zone: 0.00 (EUW)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	0.3	-6.1	2.3	2.4	-4.6	2.6	3.1	14.2	9.5	13.0	9.6	4.0	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	26.5	18.5	24.5	17.6	22.8	16.8	19.4	24.7	18.4	23.0	17.6	21.5	3.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	21.1	16.7	11.9	20.1	15.9	11.4	19.4	55.1	24.8	52.1	23.3	49.5	21.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.2	9.0	DB	-3.1	29.1	0.9	1.4	-3.8	30.1	-4.3	31.0	-4.8	31.8	-5.5	32.8	(4)
(5)			WB	-4.2	21.3	1.0	1.4	-4.9	22.3	-5.5	23.1	-6.1	23.9	-6.9	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	6.3	6.7	8.1	9.9	12.9	15.7	17.8	18.3	15.5	12.5	9.2	6.7	(6)	
	DBStd	4.97	2.82	2.75	2.60	2.55	2.69	2.38	2.31	2.53	2.03	2.46	2.61	3.05	(7)	
	HDD10.0	478	118	96	69	31	4	0	0	0	0	7	44	108	(8)	
	HDD18.3	2500	374	327	318	252	168	86	38	32	87	182	273	361	(9)	
	CDD10.0	1082	1	2	8	28	96	171	242	256	167	84	22	5	(10)	
	CDD18.3	63	0	0	0	0	1	7	22	30	3	0	0	0	(11)	
	CDH23.3	303	0	0	0	1	7	42	106	136	11	0	0	0	(12)	
	CDH26.7	52	0	0	0	0	0	7	14	30	1	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.9	4.7	4.5	4.2	4.3	3.9	3.6	3.4	3.8	4.5	3.9	4.6	(14)	
(15) Precipitation	PrecAvg	1073	113	84	84	66	66	68	68	90	98	108	118	119	(15)	
	PrecMax	1404	217	184	200	142	168	163	166	251	219	242	249	279	(16)	
	PrecMin	787	16	2	10	8	4	9	14	9	20	12	38	35	(17)	
	PrecStd	137	57	50	45	33	38	42	40	53	51	55	55	58	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.8	16.3	20.8	24.4	27.3	28.4	29.9	24.9	19.3	15.4	13.6	(19)	
		MCWB	10.6	11.2	10.4	12.9	15.9	18.0	18.6	19.6	18.0	15.2	13.5	11.9	(20)	
	2%	DB	11.8	12.2	14.2	17.4	21.9	24.1	26.2	26.8	22.4	17.7	14.5	12.8	(21)	
		MCWB	10.0	10.2	9.9	11.6	15.3	16.7	18.2	19.4	16.7	14.3	12.8	11.4	(22)	
	5%	DB	10.8	11.2	12.7	15.1	19.5	22.0	24.1	24.3	20.4	16.7	13.7	11.9	(23)	
		MCWB	9.3	9.2	9.2	10.3	14.2	15.6	17.5	18.1	15.4	14.0	12.1	10.5	(24)	
	10%	DB	10.1	10.4	11.6	13.7	17.5	20.0	22.2	22.5	19.0	15.9	13.0	11.0	(25)	
		MCWB	8.7	8.5	9.1	9.7	12.9	14.6	16.8	17.1	14.7	13.6	11.3	9.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.8	11.8	13.7	17.5	19.0	20.1	21.0	19.1	16.5	14.2	12.4	(27)	
		MCDB	12.2	13.2	13.8	19.9	22.3	25.1	26.0	27.4	23.9	18.1	14.9	13.0	(28)	
	2%	WB	10.4	10.7	10.9	12.2	15.8	17.4	19.0	19.8	17.5	15.5	13.4	11.8	(29)	
		MCDB	11.4	11.9	12.5	15.8	20.3	22.7	24.8	25.2	20.8	16.7	14.2	12.6	(30)	
	5%	WB	9.6	9.8	10.2	11.2	14.6	16.3	18.1	18.8	16.5	14.7	12.5	10.7	(31)	
		MCDB	10.6	10.9	11.7	14.1	18.8	20.6	22.6	23.1	19.1	16.2	13.5	11.6	(32)	
	10%	WB	9.0	8.9	9.6	10.3	13.4	15.4	17.2	17.8	15.8	14.0	11.6	9.7	(33)	
		MCDB	10.1	10.2	11.2	12.8	17.0	19.0	21.3	21.4	18.0	15.6	12.8	10.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.8	5.5	6.5	7.0	7.2	7.3	7.5	6.7	5.4	4.7	4.2	(35)	
		MCDBR	4.7	5.1	7.3	8.7	10.2	10.5	10.8	10.1	8.9	5.7	4.7	4.2	(36)	
	5% WB	MCWBR	4.2	4.1	4.5	4.7	5.4	5.1	5.2	4.7	4.5	3.8	3.9	4.2	(37)	
		MCWBR	4.5	4.7	5.6	7.4	9.3	9.0	9.0	9.2	7.4	4.9	4.3	4.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.331	0.345	0.377	0.397	0.402	0.393	0.391	0.377	0.369	0.355	0.330	0.321	(40)	
		taud	2.346	2.372	2.287	2.254	2.281	2.341	2.350	2.406	2.422	2.425	2.400	2.323	(41)	
	Ebn at Noon		670	773	811	839	852	863	857	852	816	751	670	613	(42)	
		Edh at Noon	64	82	106	123	125	119	116	105	92	77	60	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.78	1.48	2.61	4.02	4.94	5.36	5.02	4.16	3.13	1.77	0.94	0.61	(44)	
	RadStd		0.07	0.18	0.33	0.46	0.25	0.57	0.47	0.37	0.23	0.16	0.09	0.08	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page